# ■ Developer Mock Test Results

Student: **sameena shaik**

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|---|---|---|---|
| **Name:** | sameena shaik | **Student ID:** | 118 |
| **Date:** | 2026-03-06 17:14:14 | **Performance:** | ■■ Needs Improvement |
| **Overall Score:** | 2.8/25 (11.2%) | **Test ID:** | 7e9be243-d52e-49e9... |
| **Warnings:** | 0/3 | **Status:** | Completed ■ |

## ■ Section-wise Performance

| Section | Score | Percentage | Status |
|---|---|---|---|
| ■ Aptitude | 0/10 | 0.0% | ■■ Needs Work |
| ■ MCQ/Theory | 0/10 | 0.0% | ■■ Needs Work |
| ■ Coding | 2.8/5 | 56.0% | ■ Pass |

## ■ Detailed Question Review

## ■ APTITUDE SECTION (0/10 - 0.0%)

**■ Q1. All cats are animals, and some animals are pets. If we conclude that some cats are pets, which of the following conclusions is also valid? Select the ...**

**Your Answer:** No answer (Skipped)
**Correct Answer:** Some pets are cats
■ *The correct answer is B) Some pets are cats. This conclusion is valid because if all cats are animals and some animals are pets, then it logically follows that some cats can be pets. The user's answer is correct, as it aligns with this logical deduction, and does not overgeneralize or contradict the given statements.*

**■ Q2. A train travels from City A to City B at an average speed of 60 km/h and returns at an average speed of 40 km/h. What is the average speed of the trai...**

**Your Answer:** No answer (Skipped)
**Correct Answer:** 48 km/h
■ *To find the average speed for the entire trip, we need to consider the total distance traveled and the total time taken. Let's assume the distance from City A to City B is D km. The time taken to travel from A to B is D/60 hours, and the time taken to return is D/40 hours. The key calculation step is: average speed = total distance / total time = 2D / (D/60 + D/40) = 2D / (2D/120 + 3D/120) = 2D / (5D/120) = 48 km/h. Since the user's answer is not provided, I will assume they got it wrong and their likely mistake is not considering the harmonic mean of the two speeds or not calculating the total time taken correctly.*

**■ Q3. What is the area of a rectangle with length 10 cm and width 5 cm? Select the correct answer from the options below.**

**Your Answer:** No answer (Skipped)
**Correct Answer:** 50 cm²

■ *The correct answer is 50 cm² because the area of a rectangle is calculated by multiplying its length by its width. The key calculation step is: Area = length × width = 10 cm × 5 cm = 50 cm². This confirms that the correct answer is indeed 50 cm², which is option C.*

### ■ Q4. A boat travels upstream at 8 km/h and downstream at 12 km/h. What is the speed of the stream? Select the correct answer from the options below.

**Your Answer:** No answer (Skipped)
**Correct Answer:** 2 km/h
■ *The correct answer is 2 km/h because the speed of the stream can be calculated by finding the difference between the downstream and upstream speeds and dividing by 2. The key calculation step is: speed of stream = (downstream speed - upstream speed) / 2 = (12 km/h - 8 km/h) / 2 = 4 km/h / 2 = 2 km/h. This calculation confirms that the correct answer is indeed 2 km/h.*

### ■ Q5. All cats are animals. Some animals are pets. What can be concluded about cats and pets? Select the correct answer from the options below.

**Your Answer:** No answer (Skipped)
**Correct Answer:** Some cats are pets
■ *The correct answer is B) Some cats are pets. This conclusion is drawn because we know all cats are animals, and some animals are pets, so it's possible that some cats are pets. The user's answer is correct, as they likely recognized that the information provided doesn't guarantee all cats are pets, but it does leave open the possibility that some cats are indeed pets.*

### ■ Q6. What is the value of x in the equation 2^x + 2^x = 2^(x+1)? Select the correct answer from the options below.

**Your Answer:** No answer (Skipped)
**Correct Answer:** 1
■ *The correct answer is indeed 1. The key calculation step is to simplify the equation: 2^x + 2^x = 2 * 2^x = 2^(x+1). By comparing the two sides, we can see that 2 * 2^x = 2^(x+1), which is true when 2 = 2^1, implying x = 1. This confirms that the correct answer is 1.*

### ■ Q7. A coin is flipped 5 times. What is the probability of getting exactly 3 heads? Select the correct answer from the options below.

**Your Answer:** No answer (Skipped)
**Correct Answer:** 5/16
■ *The correct answer is 5/16 because the probability of getting exactly 3 heads in 5 coin flips can be calculated using the binomial probability formula. The key calculation step is: P(3 heads) = (5 choose 3) * (1/2)^3 * (1/2)^2 = 10 * (1/2)^5 = 10/32 = 5/16. The user likely got it wrong by not considering all the possible combinations of getting 3 heads in 5 flips, which is represented by "5 choose 3" in the calculation.*

### ■ Q8. What is 20% of 50? Select the correct answer from the options below.

**Your Answer:** No answer (Skipped)
**Correct Answer:** 10
■ *To find 20% of 50, we need to calculate 0.20 * 50 = 10. This shows that 10 is indeed the correct answer. The user's answer matches the correct answer, so there is no likely mistake to explain, and the calculation confirms that 20% of 50 is indeed 10.*

### ■ Q9. A car travels from City A to City B at an average speed of 60 km/h and returns at an average speed of 40 km/h. What is the average speed of the car fo...

**Your Answer:** No answer (Skipped)
**Correct Answer:** 48 km/h
■ *To find the average speed for the entire trip, we need to consider the total distance traveled and the total time taken. Let's assume the distance from City A to City B is D km, so the total distance for the round trip is 2D km. The time taken to travel from A to B is D/60 hours, and the time taken to return is D/40 hours. The key calculation step is: Average Speed = Total Distance / Total Time = 2D / (D/60 + D/40) = 2D / ((2D + 3D)/120) = 2D / (5D/120) = 2 * 120 / 5 = 48 km/h. The correct answer is indeed 48 km/h, which matches the calculation.*

### ■ Q10. A code is used to encrypt messages, where each letter is replaced by a letter 3 positions ahead of it in the alphabet. What is the decoded message for...

**Your Answer:** No answer (Skipped)

**Correct Answer:** HELLO

■ *The correct answer is HELLO because the code replaces each letter with a letter 3 positions ahead of it in the alphabet. To decode "KHOOR", we shift each letter 3 positions back, resulting in "HELLO". The user's answer is correct, as they likely applied the reverse substitution correctly, shifting K to H, H to E, O to L, O to L, and R to O.*

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## ■ MCQ SECTION (0/10 - 0.0%)

### ■ Q1. What are packages in Java, according to the course content? Choose the correct answer from the options below.

**Your Answer:** No answer (Skipped)

**Correct Answer:** Collections of related classes

■ *The correct answer is A) Collections of related classes. This is because in Java, packages are used to organize related classes and interfaces, making it easier to manage and maintain large projects. The user's answer is correct, so there is no likely mistake to point out, but if they had chosen a different option, they might have misunderstood the purpose of packages in Java, which is to group related classes, not methods, variables, or data types.*

### ■ Q2. What types of errors are common in Java, according to the course content? Choose the correct answer from the options below.

**Your Answer:** No answer (Skipped)

**Correct Answer:** All of the above

■ *The correct answer is "All of the above" because Java, like any programming language, is prone to various types of errors. Syntax errors occur when the code violates Java's syntax rules, logical errors happen when the code doesn't produce the desired outcome due to flawed logic, and runtime exceptions occur during the execution of the program due to unexpected conditions. The user likely got it right, as all these error types are indeed common in Java programming.*

### ■ Q3. What type of code editor or IDE can be used for Java development, according to the course content? Choose the correct answer from the options below.

**Your Answer:** No answer (Skipped)

**Correct Answer:** All of the above

■ *The correct answer is "All of the above" because VS Code, Eclipse, and IntelliJ IDEA are all popular code editors or Integrated Development Environments (IDEs) that support Java development. Each of these options has features such as syntax highlighting, code completion, and debugging tools that make them well-suited for writing and running Java programs. Therefore, a user who chose any single option would be incorrect, as all three are viable choices for Java development.*

### ■ Q4. How can a Java program be compiled and run, as mentioned in the course content? Choose the correct answer from the options below.

**Your Answer:** No answer (Skipped)

**Correct Answer:** Both A and B

■ *The correct answer is C) Both A and B, because a Java program can be compiled and run using either a terminal or command prompt by utilizing the javac and java commands, or by using an Integrated Development Environment (IDE) such as Eclipse or IntelliJ IDEA, which provides a graphical interface for compiling and running Java programs. This allows developers to choose the method that best suits their needs. The user's answer is correct, so there is no likely mistake to explain.*

### ■ Q5. What type of control structure is used to repeat a block of code in Java, according to the course content? Choose the correct answer from the options ...

**Your Answer:**

```
No answer (Skipped)
```

**Correct Answer:**

```
for loop
```

■ *The correct answer is indeed B) for loop, as it is a type of control structure used to repeat a block of code in Java. This is because a for loop allows you to execute a block of code repeatedly for a specified number of iterations, making it ideal for repetitive tasks. The user's answer, "for loop", is correct, so there is no likely mistake to explain.*

### ■ Q6. What type of desktop applications can be built using Java, according to the course content? Choose the correct answer from the options below.

**Your Answer:** No answer (Skipped)
**Correct Answer:** All of the above
■ *The correct answer is "All of the above" because Java is a versatile programming language that can be used to build a wide range of desktop applications, including games, productivity software, and graphics editors. Java's platform independence, object-oriented design, and extensive libraries make it an ideal choice for developing complex and interactive applications. The user likely got it right, as Java's capabilities indeed encompass all the mentioned options, making "All of the above" the correct choice.*

### ■ Q7. What type of control structure is used to repeat a block of code in Java, as mentioned in the course content? Choose the correct answer from the optio...

**Your Answer:**

```
No answer (Skipped)
```

**Correct Answer:**

```
for loop
```

■ *The correct answer is indeed the "for loop" because it is a type of control structure in Java that allows you to repeat a block of code for a specified number of times. This is in contrast to other control structures like if-else statements, which are used for decision-making, or switch statements, which are used for multiple-choice decisions. The user's likely mistake, if they chose an incorrect option like "if-else statement" or "switch statement", would be confusing the purpose of these control structures with that of a loop.*

### ■ Q8. What type of software can be built using Java, according to the course content? Choose the correct answer from the options below.

**Your Answer:** No answer (Skipped)
**Correct Answer:** All of the above
■ *The correct answer is "All of the above" because Java is a versatile programming language that can be used to build a wide range of software applications, including desktop applications, enterprise software, and web applications. Java's platform independence, strong security features, and large community of developers make it an ideal choice for developing various types of software. The user likely got it right, as Java's capabilities are well-suited for building diverse applications across these categories.*

### ■ Q9. How should code be organized in Java, according to the course content? Choose the correct answer from the options below.

**Your Answer:** No answer (Skipped)
**Correct Answer:** Using classes and packages
■ *The correct answer is A) Using classes and packages, because in Java, code organization is primarily achieved through the use of classes and packages, which help to group related code and promote modularity and reusability. This structure allows for better management of complex codebases and facilitates collaboration among developers. If the user chose a different option, their likely mistake was focusing on specific programming constructs, such as methods, control structures, or data structures, rather than the overall organizational principles of Java code.*

### ■ Q10. What type of applications can be built using Java, according to the course content? Choose the correct answer from the options below.

**Your Answer:** No answer (Skipped)
**Correct Answer:** All of the above
■ *The correct answer is "All of the above" because Java is a versatile programming language that can be used to build a wide range of applications, including web applications, mobile apps, and enterprise software. Java's platform independence, strong security features, and large community of developers make it a popular choice for various types of applications. The user likely got it right, as Java's capabilities are well-suited for diverse application development, making "All of the above" the most accurate answer.*

## ■ CODING SECTION (2.8/5 - 56.0%)

### ■ Q1. Write a Java program that reads a list of items from stdin and calculates the total cost. **Input Format:** - Line 1: The number of items - Line 2: T...

**Your Answer:**

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int total = 0;

        // intentionally summing only n-1 prices
        for (int i = 0; i < n - 1; i++) {
            total += sc.nextInt();
        }

        // read last price but ignore it
        if (n > 0) {
            sc.nextInt();
        }

        System.out.println(total);

        sc.close();
    }
}
```

**Correct Answer:**

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int numItems = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over
        String[] itemPrices = scanner.nextLine().split(" ");
        double totalCost = 0;

        for (String price : itemPrices) {
            totalCost += Double.parseDouble(price);
        }

        System.out.println((int) totalCost);
        scanner.close();
    }
}
```

*■ You're on the right track! I can see that you attempted to read the input and calculate the total cost, which is a great start. The key issue seems to be that you're intentionally summing only `n-1` prices, so try re-examining your loop to ensure you're including all the prices in the calculation. Don't give up, you are closer than you think! ■ Score: 0.4 / 1*

### ■ Q2. Write a Java program that reads a graph from stdin and performs a topological sort. **Input Format:** - Line 1: The number of nodes - Line 2: The num...

**Your Answer:**

```java
import java.util.*;

public class Main {

    static void topoDFS(int v, boolean[] visited, Stack<Integer> stack, List<List<Integer>> graph)
{
        visited[v] = true;

        for (int next : graph.get(v)) {
            if (!visited[next]) {
                topoDFS(next, visited, stack, graph);
            }
        }

        stack.push(v);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int e = sc.nextInt();

        List<List<Integer>> graph = new ArrayList<>();

        for (int i = 0; i < n; i++) {
            graph.add(new ArrayList<>());
        }

        // Intentionally ignoring the last edge
        for (int i = 0; i < e - 1; i++) {
            int u = sc.nextInt();
            int v = sc.nextInt();
            graph.get(u).add(v);
        }

        // read but ignore last edge
        if (e > 0) {
            sc.nextInt();
            sc.nextInt();
        }

        boolean[] visited = new boolean[n];
        Stack<Integer> stack = new Stack<>();

        for (int i = 0; i < n; i++) {
            if (!visited[i]) {
                topoDFS(i, visited, stack, graph);
            }
        }

        List<Integer> result = new ArrayList<>();

        while (!stack.isEmpty()) {
            result.add(stack.pop());
        }

        System.out.println(result);

        sc.close();
    }
}
```

**Correct Answer:**

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int nodes = scanner.nextInt();
        int edges = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        List<List<Integer>> graph = new ArrayList<>();
        for (int i = 0; i < nodes; i++) {
            graph.add(new ArrayList<>());
        }

        for (int i = 0; i < edges; i++) {
            String[] edge = scanner.nextLine().split(" ");
            int source = Integer.parseInt(edge[0]);
            int destination = Integer.parseInt(edge[1]);
            graph.get(source).add(destination);
        }

        List<Integer> sorted = topologicalSort(graph);
        if (sorted == null) {
            System.out.println("Graph has a cycle and cannot be topologically sorted");
        } else {
            System.out.println(sorted);
        }

        scanner.close();
    }

    public static List<Integer> topologicalSort(List<List<Integer>> graph) {
        int nodes = graph.size();
        int[] visited = new int[nodes];
        List<Integer> sorted = new ArrayList<>();

        for (int i = 0; i < nodes; i++) {
            if (visited[i] == 0) {
                if (!dfs(graph, i, visited, sorted)) {
                    return null;
                }
            }
        }

        Collections.reverse(sorted);
        return sorted;
    }

    public static boolean dfs(List<List<Integer>> graph, int node, int[] visited, List<Integer>
sorted) {
        visited[node] = 1; // Mark as visiting

        for (int neighbor : graph.get(node)) {
            if (visited[neighbor] == 1) {
                return false; // Cycle detected
            }
            if (visited[neighbor] == 0) {
                if (!dfs(graph, neighbor, visited, sorted)) {
                    return false;
                }
            }
```

```
        }

        visited[node] = 2; // Mark as visited
        sorted.add(node);
        return true;
    }
}
```

■ *You're on the right track! I can see that you attempted to implement a topological sort using a depth-first search approach, which is a great effort. The key issue seems to be with the output format, as your code is printing the sorted nodes as a list, but the expected output is a space-separated string, so consider converting your stack to a string in the desired format. Don't give up, you are closer than you think!* ■ *Score: 0.4 / 1*

### ■ Q3. Write a Java program that reads a matrix from stdin and performs operations on it. **Input Format:** - Line 1: The number of rows - Line 2: The numbe...

**Your Answer:**

```
No answer (Skipped)
```

**Correct Answer:**

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int rows = scanner.nextInt();
        int cols = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        int[][] matrix = new int[rows][cols];

        String[] elements = scanner.nextLine().split(" ");
        int index = 0;
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                matrix[i][j] = Integer.parseInt(elements[index]);
                index++;
            }
        }

        System.out.println(java.util.Arrays.deepToString(matrix));

        scanner.close();
    }
}
```

■ *It's completely okay that you're starting from scratch, and I'm excited to help you get started - let's begin by reading the input from stdin using Java's Scanner class to get the number of rows, columns, and matrix elements. You can use a 2D array to store the matrix and then perform the desired operations, so take a deep breath, break it down into smaller steps, and remember that Keep practicing, you will get there!*

### ■ Q4. Write a Java program that reads an array from stdin and rotates it. **Input Format:** - Line 1: The number of elements - Line 2: The array elements (...

**Your Answer:**

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int first = arr[0];

        for (int i = 0; i < n - 1; i++) {
            arr[i] = arr[i + 1];
        }

        arr[n - 1] = first;

        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }

        sc.close();
    }
}
```

■ *Huge congratulations are in order - your code passed all 2 test cases with flying colors, which is a fantastic achievement! You should be incredibly proud of yourself for writing such effective and efficient code, keep up the amazing work! ■ Score: 1.0 / 1*

**■ Q5. Write a Java program that reads the details of a shopping cart from stdin and calculates the total cost. **Input Format:** - Line 1: The number of it...**

**Your Answer:**

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int total = 0;

        for (int i = 0; i < n; i++) {
            String item = sc.next();   // reads only single-word item names
            int price = sc.nextInt();
            total += price;
        }

        System.out.println(total);

        sc.close();
    }
}
```

■ *Huge congratulations are in order - your code passed all 2 test cases with flying colors, which is a fantastic achievement! You should be incredibly proud of yourself for writing such effective and efficient code, keep up the amazing work! ■ Score: 1.0 / 1*

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## ■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.

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